Organization of Gene

“Structure and Function of DNA & RAN”

SGL - 3

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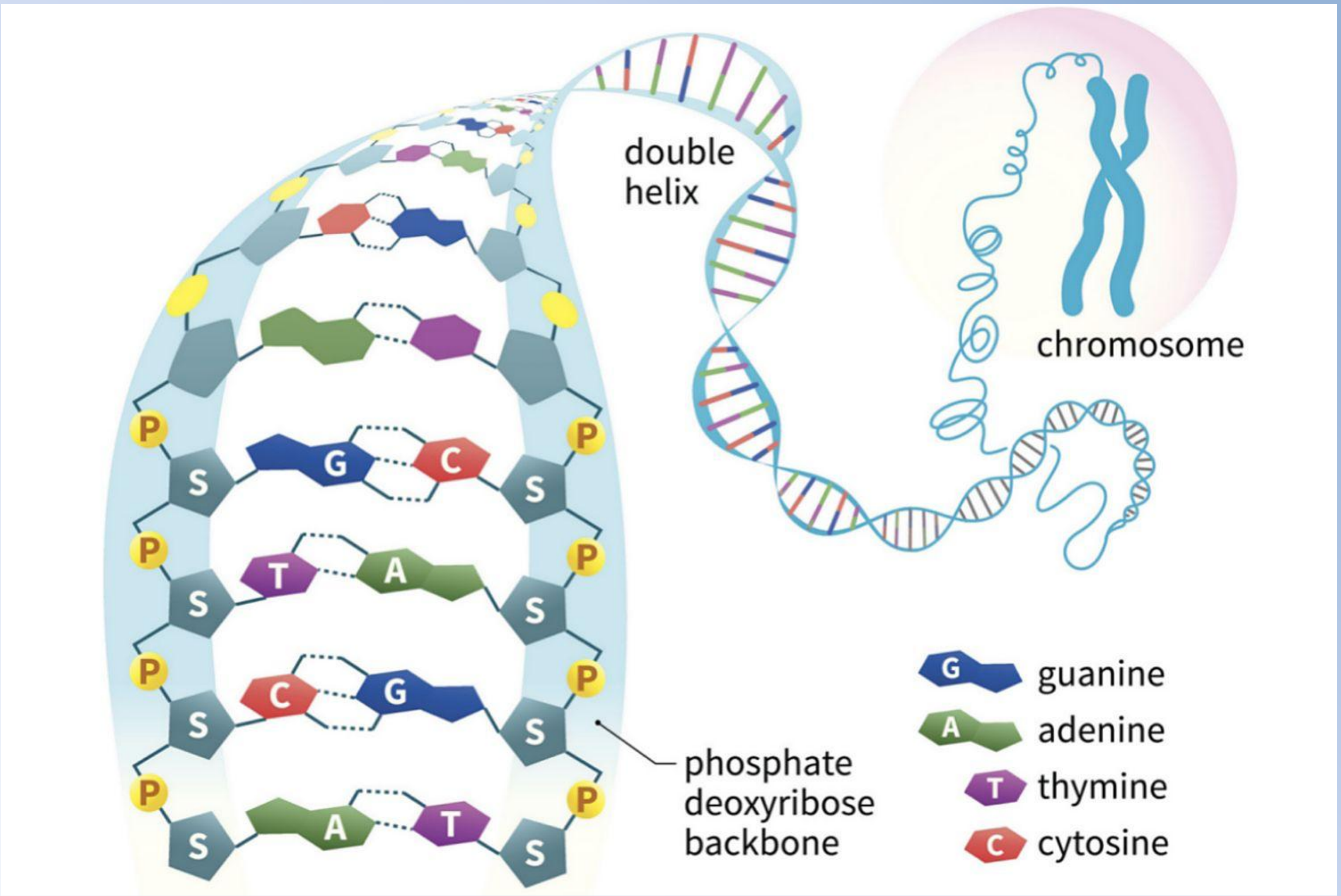
Molecular Biology :

DNA Structure: **Quick Review**



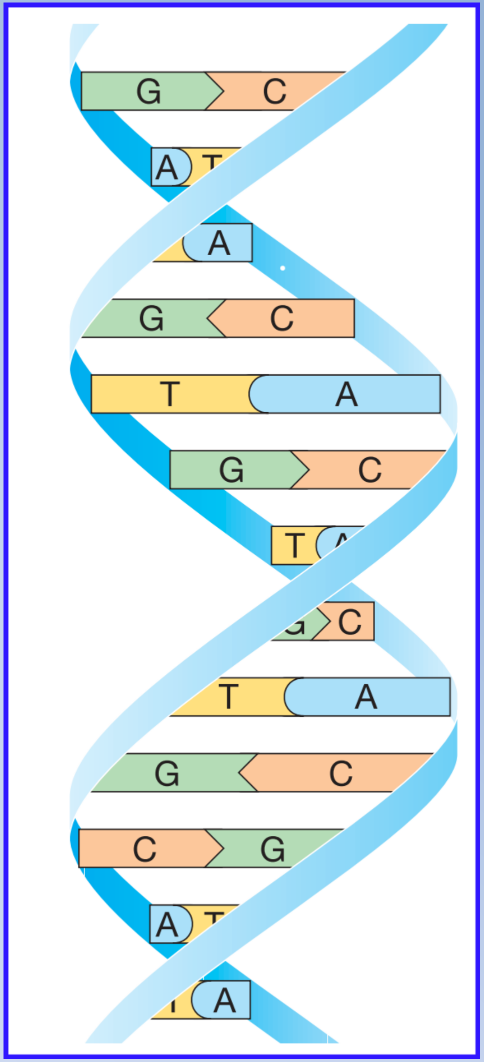
Molecular Biology :

DNA Feature: ***Quick Review***

DNA as a double helix

- The double helix is the characteristic structural feature of DNA. The two helical polynucleotide chains are wound around each other along a common axis. The nucleotide base pairs (bp), either A–T or G–C, lie within.

- Because of the fixed spatial relationship of the nucleotide bases within the double helix and opposite each other, the two chains of the double helix are exactly complementary.

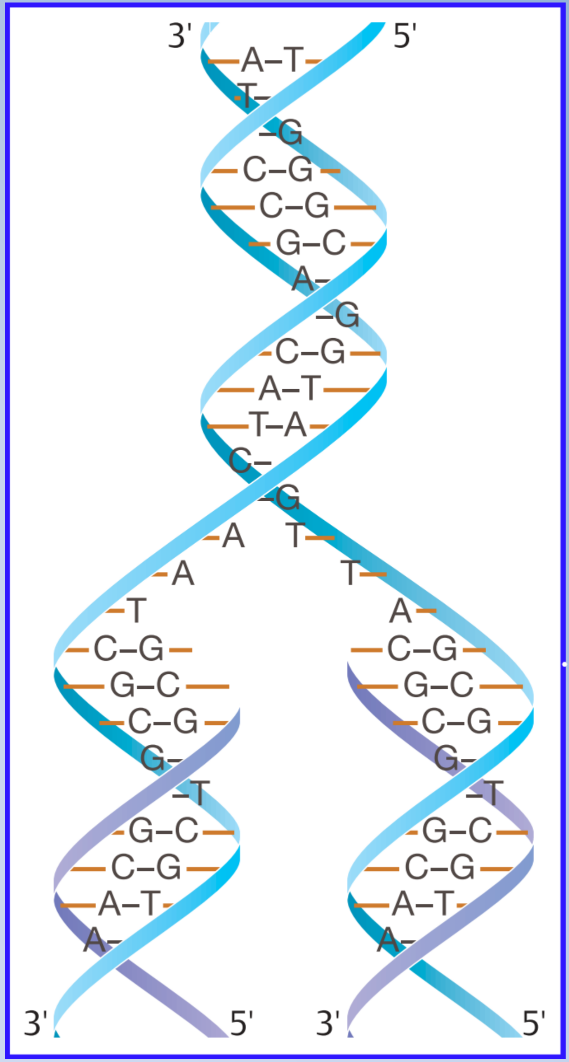


Molecular Biology :

DNA Replication

- Since the nucleotide chains lying opposite each other within the double helix are strictly complementary, each can serve as a pattern (template) for the formation (replication) of a new chain when the helix is opened.

- DNA replication is semiconservative, i.e., one completely new strand will be formed and one strand retained



The Main Characteristics of DNA:

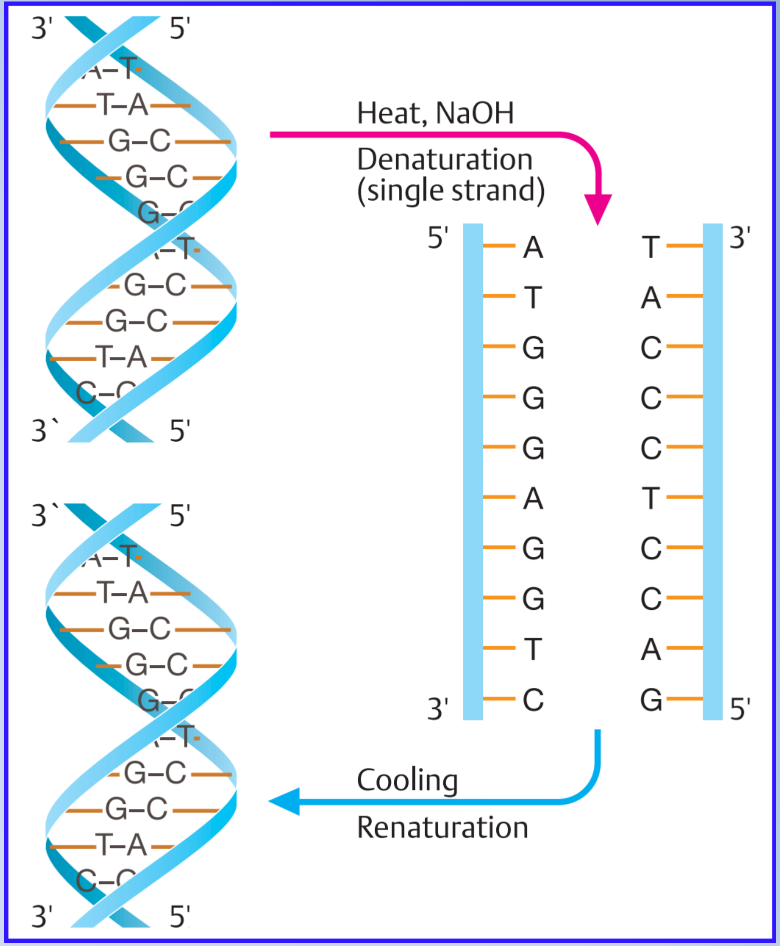
A- DNA Denaturation and renaturation

- The noncovalent hydrogen bonds between the nucleotide base pairs are weak, but DNA is stable at physiological temperatures because it is a very long molecule with presence of many other factors that maintains DNA stability.

- The two complementary strands can be separated *(DE_NATURATION)* by means of relatively weak chemical reagents (e.g., alkali, formamide, or urea) or by careful heating. The resulting single-stranded molecules are relatively stable.

- With cooling, complementary single strands can reunite to form double-stranded molecules *(RE_NATURATION).*

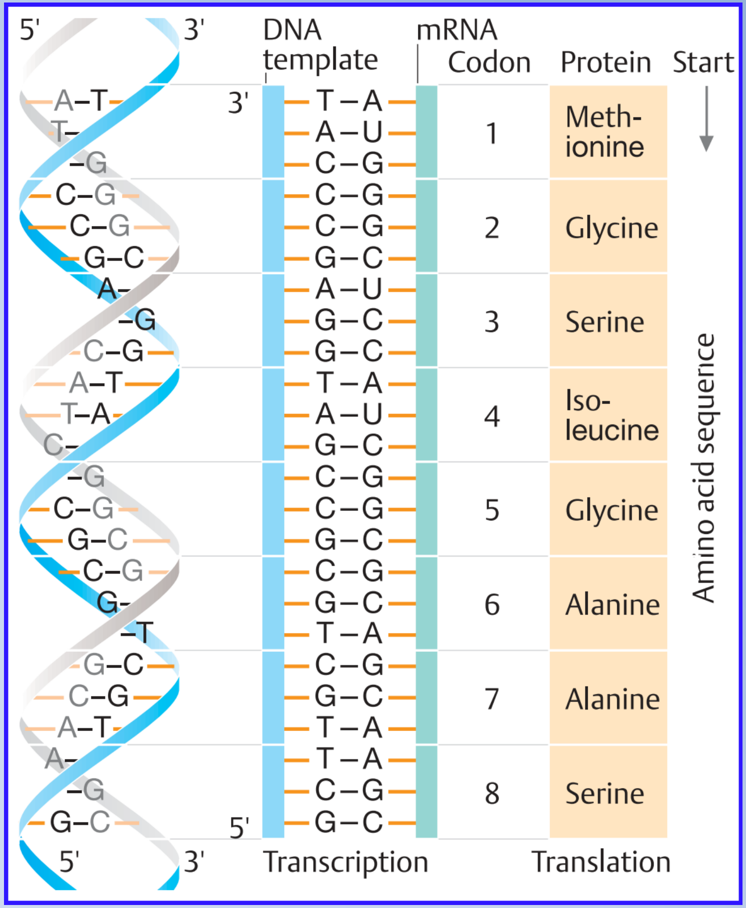
- Noncomplementary single strands do not unite



The Main Characteristics of DNA:

Transmission of genetic information

- Genetic information lies in the sequence of nucleotide base pairs (A–T or G–C). A sequence of three base pairs represents a code word (codon) for an amino acid.
- The codon sequence determines a corresponding sequence of amino acids. These form a polypeptide (gene product).
- The sequence of the nucleotide bases is first transferred *(TRANSCRIPTION)* from one DNA strand to a further information-bearing molecule (mRNA, messenger RNA).
- Then the nucleotide base sequence of the mRNA serves as a template for a sequence of amino acids corresponding to the order of the codons *(TRANSLATION).*



Most Significant Features of DNA:

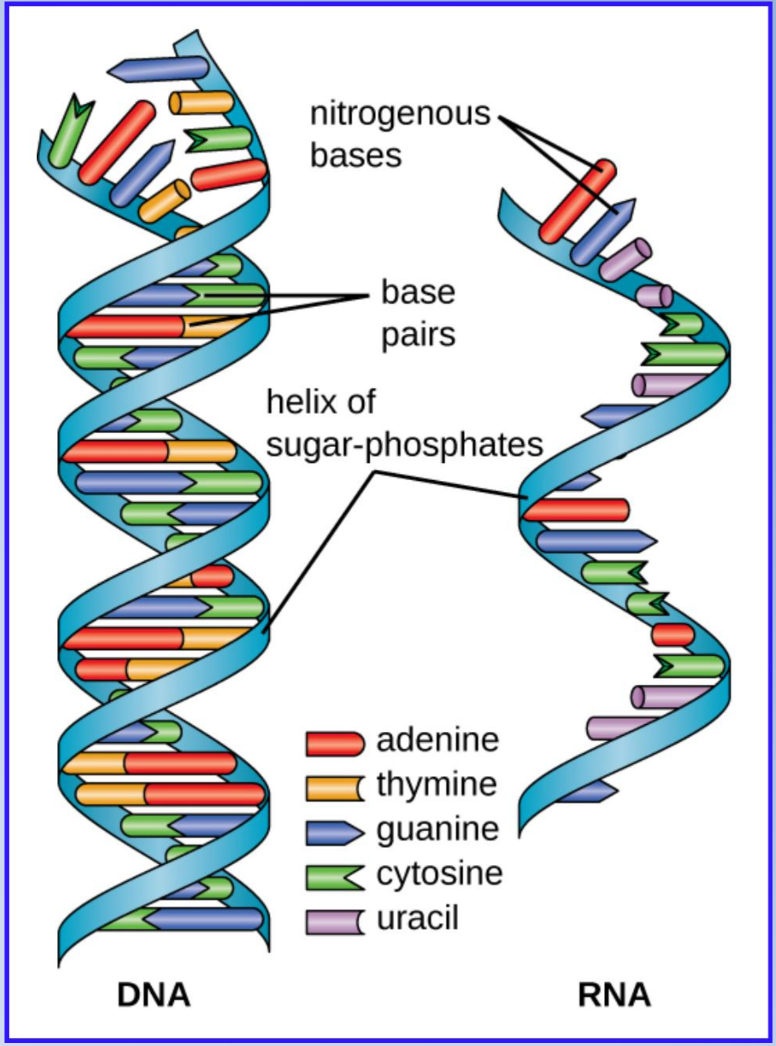
- The DNA molecule is a double helix structure.
- The two strands of a DNA molecule are anti-parallel(one is 3' to 5' direction and another one is 5' to 3' direction) and twisted.
- It is made up of four nucleotides - (A), (G), (C), & (T).
- Each nucleotide consists of a phosphate group, a pentose sugar (deoxyribose sugar in case of DNA), and a nitrogen base.
- The two strands are attached with hydrogen bonds in which there are two hydrogen bonds between A and T and three hydrogen bonds between C and G.
- It has complementary base pairing which means one molecule of the strands chemically matches the sequences on another strand.
- DNA is a molecule that carries all the genetic information for the function and development of an organism.
- Eukaryotes are (diploid); contain two copies of each chromosome so it contains two copies of each gen. Prokaryotes are typically (haploid), it usually having a single circular chromosome contains only one copy of each gene.
- The nucleotides in DNA code for protein.
- DNA molecules carry genes that store all the genetic information.

Molecular Biology : *Ribonucleic Acid (RNA):*

RNA Characteristic:

Ribonucleic acid (RNA) is a molecule that is present in the majority of living organisms and viruses. Structurally speaking, ribonucleic acid (RNA), is quite similar to DNA. whereas

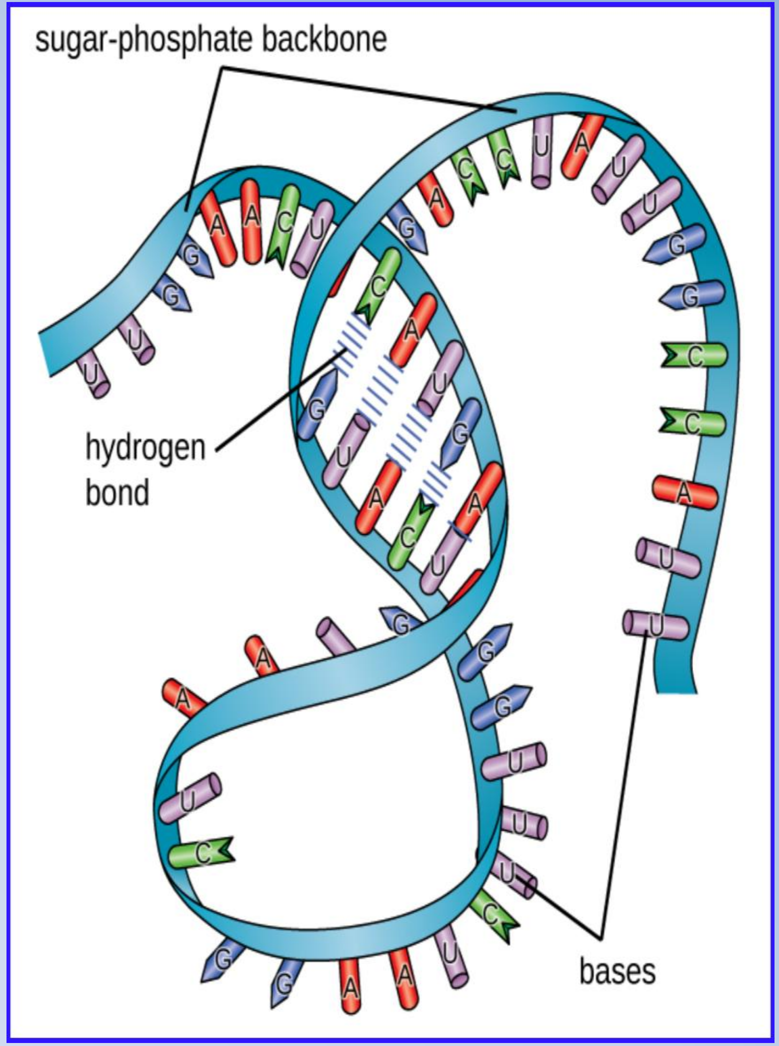
- DNA molecules are typically long and double stranded, RNA molecules are much shorter and are typically single stranded, but there are special RNA viruses that are double-stranded.
- RNA molecules perform a variety of roles in the cell but are mainly involved in the process of protein synthesis and its regulation.



Molecular Biology : *Ribonucleic Acid (RNA):*

RNA Characteristic:

- The means of RNA synthesis and the way that it functions differs between eukaryotes and prokaryotes.
- RNA does serve as the hereditary information and genetic material in RNA viruses
- The RNA-specific pyrimidine uracil (U) forms a complementary base pair with (A) and is used instead of the (T) used in DNA.
- Even though RNA is single stranded, most types of RNA molecules show extensive intramolecular base pairing between complementary sequences within the RNA strand, creating a predictable three-dimensional structure essential for their function.

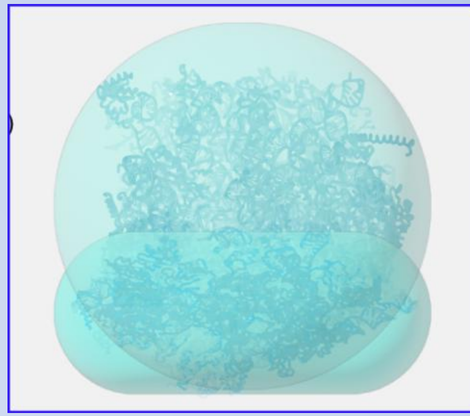


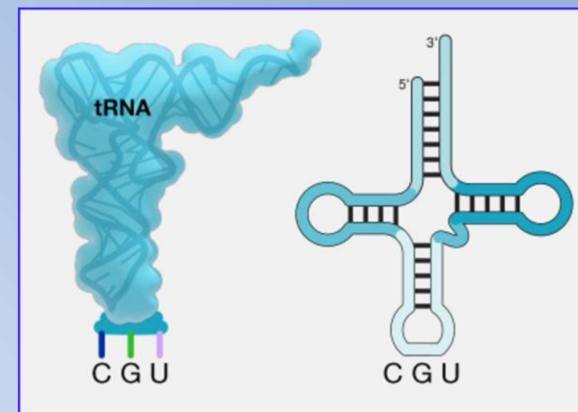
Molecular Biology : *Ribonucleic Acid (RNA):*

RNA Types & Functions:

The three main types of RNA directly involved in protein synthesis are

- messenger RNA (mRNA),
- ribosomal RNA (rRNA),
- transfer RNA (tRNA)





	mRNA	rRNA	tRNA
Structure	Short, unstable, single-stranded RNA corresponding to a gene encoded within DNA	Longer, stable RNA molecules composing 60% of ribosome's mass	Short (70-90 nucleotides), stable RNA with extensive intramolecular base pairing; contains an amino acid binding site and an mRNA binding site
Function	Serves as intermediary between DNA and protein synthesis; used by ribosome to direct synthesis of protein it encodes	Ensures the proper alignment of mRNA, tRNA, and ribosome during protein synthesis; catalyzes peptide bond formation between amino acids	Each amino acid has its own tRNA and carries the correct amino acid to the site of protein synthesis in the ribosome

Molecular Biology : *Fundamentals*

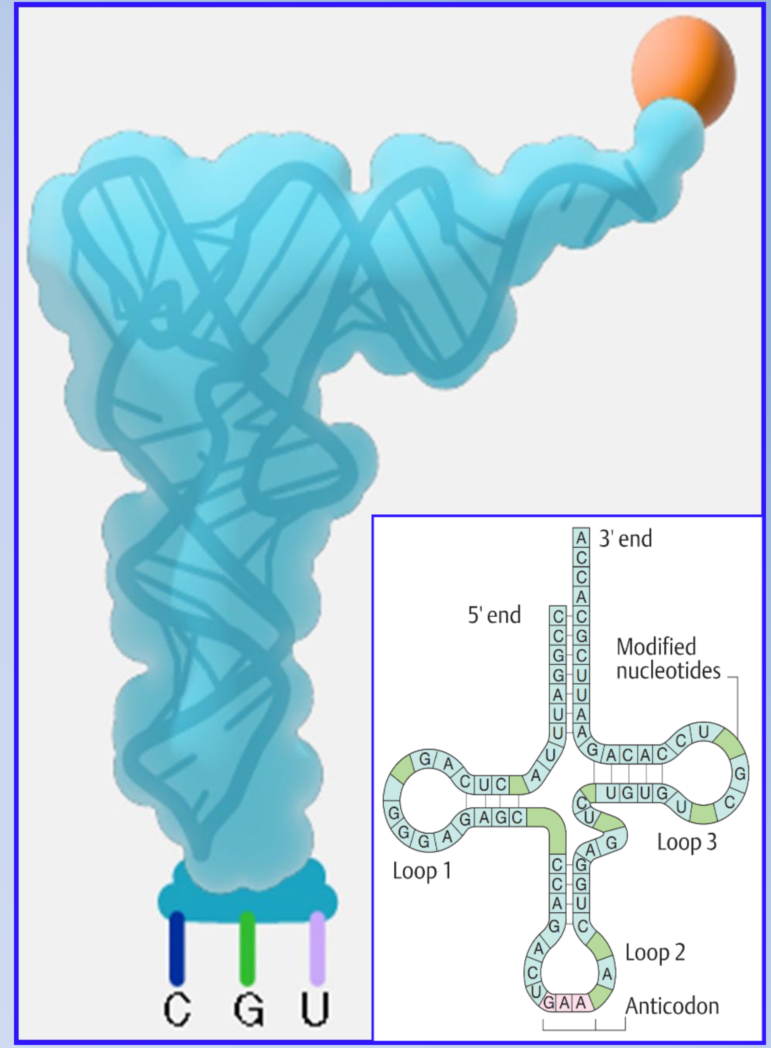
Ribonucleic Acid (RNA): Characteristic of tRNA

Transfer RNA has a characteristic, cloverleaf-like structure. illustrated here by yeast phenylalanine tRNA

(1) It has three single-stranded loop regions and four double-stranded "stem" regions.

(2) The three-dimensional structure is complex, but various functional areas can be differentiated, such as the recognition site (anticodon) for the mRNA codon and the binding site for the respective amino acid (acceptor stem) on the 3! end (acceptor end).

(3) Each amino acid has its own tRNA, which has a region that is complementary to its codon of the mRNA (anticodon).



DNA Packing & and supercoiling:

- The length of a chromosome greatly exceeds the length of the cell, so a chromosome needs to be packaged into a very small space to fit within the cell.

- The combined length of all of the 3 billion base pairs of DNA of the human genome would measure approximately 2 meters if completely stretched out, and some eukaryotic genomes are many times larger than the human genome.

- DNA supercoiling refers to the process by which DNA is twisted to fit inside the cell.

- Proteins known to be involved in supercoiling include topoisomerases; these enzymes help maintain the structure of supercoiled chromosomes, preventing overwinding of DNA during certain cellular processes like DNA replication.

Molecular Biology : *Fundamentals*

DNA Packing & and supercoiling:

- During DNA packaging, DNA-binding proteins called histones perform various levels of DNA wrapping and attachment to scaffolding proteins.

- Histones are proteins with a high proportion of positively charged amino acids (lysine and arginine), which enable them to bind firmly to the negatively charged DNA double helix.

- In eukaryotes, the packaging of DNA by histones may be influenced by environmental factors that affect the presence of methyl groups on certain cytosine nucleotides of DNA.

- The influence of environmental factors on DNA packaging is called epigenetics. Epigenetics is another mechanism for regulating gene expression without altering the sequence of nucleotides.